Comparative Table: AE vs VAE vs DCGAN vs Stable Diffusion

Assignment-02 | Generative AI Models | CO3 (reconstruction: AE, VAE) and CO4 (generation: DCGAN, Stable Diffusion). Bracketed numbers are clickable links to the references on page 2.

Criterion	Autoencoder CO3 | reconstruction	VAE CO3 | reconstruction + sampling	DCGAN CO4 | generation (GAN)	Stable Diffusion CO4 | generation (latent diffusion)
1. Working principle	• Encoder **squeezes** the image into a small latent code; decoder **rebuilds** it. • Trained only to minimise reconstruction error (MSE in our run). • Each image maps to a single fixed point in latent space.	• Encoder outputs a **mean and variance**, i.e. a Gaussian per image [1]. • A code is sampled with the **reparameterisation trick**, then decoded. • Loss = reconstruction + **KL divergence** to N(0, I), which shapes the latent space.	• A **generator** turns noise into images; a **discriminator** separates real from fake [2]. • The two are trained against each other (minimax game). • Convolutional design rules make this stable enough for images [3].	• Starts from random **noise in a compressed latent space** and removes it step by step [4, 5]. • Trained to reverse gradual noising; the text prompt steers every step. • A VAE decoder turns the final latent into pixels.
2. Architecture and components	• Conv encoder, **16-value bottleneck**, transposed-conv decoder. • About **0.18 M** parameters (our notebook). • Deterministic: same input always gives same code.	• Same decoder as the AE; encoder ends in **two heads** (mean, log-variance) plus a **sampling layer**. • About **0.53 M** parameters (our notebook). • Same 16-value latent size for a fair comparison.	• **Generator:** dense layer, transposed convolutions, BatchNorm, tanh output. • **Discriminator:** strided convolutions, LeakyReLU, no pooling [3]. • Generator about **2.33 M** parameters (our notebook).	• **CLIP-based text encoder**, **U-Net denoiser** with cross-attention, **VAE** (latent to pixels), **noise scheduler** [5]. • Hundreds of millions of parameters in total; diffusion runs in latent space, not pixel space.
3. Advantages	• Simple and **fast to train**. • Good for compression, denoising, feature learning and anomaly detection. • Stable, easy to debug.	• **Stable training**. • **Smooth latent space**: sampling and interpolation work. • Gives both reconstruction and generation from one model.	• **Sharper** images than a VAE. • Generates in a **single forward pass** (fast sampling). • Learns useful convolutional features.	• **Highest realism and detail**. • **Text control** over content and style. • Diverse outputs and stable training; open pretrained weights.
4. Limitations	• Unstructured latent space: **cannot generate** new images. • Detail is lost through the bottleneck; smoothed textures. • Can learn a near-identity mapping if the bottleneck is too large.	• **Blurrier** than GAN or diffusion output. • KL term trades sharpness for latent structure. • Risk of posterior collapse if the KL term dominates.	• **Unstable training**, sensitive to hyper-parameters. • **Mode collapse** (repeated, low-variety samples). • No direct control over the output; hard to evaluate.	• **Slow**: many denoising steps per image. • Heavy GPU memory and compute. • Artifacts (hands, text, counting); **inherits dataset bias**.

Criterion	**Autoencoder** CO3 | reconstruction	**VAE** CO3 | reconstruction + sampling	**DCGAN** CO4 | generation (GAN)	**Stable Diffusion** CO4 | generation (latent diffusion)
5. Applications	• Image compression • Denoising • Dimensionality reduction • Anomaly detection	• Sample generation • Latent interpolation • Representation learning • Anomaly detection	• Image synthesis • Data augmentation • Face and style generation • Unsupervised feature learning	• Text-to-image • Inpainting and image editing • Super-resolution • Design and concept art
6. Differences in outputs (our runs)	• Close copies of simple shapes with **smoothed texture** (mean MSE **0.0108**). • Random codes decode to **unrecognisable blobs**.	• **Slightly blurrier** than AE (mean MSE **0.0145**). • Random codes decode to **recognisable garments**.	• **Sharper** new 28x28 samples, but some are **malformed**. • No prompt control.	• **High-resolution colour** output that follows the prompt. • Slowest per image.

At a glance: scorecard, key numbers and when to use which

Ratings are qualitative (5 = best), based on our notebook runs and the cited papers. They are for quick comparison, not measured scores.

Quality dimension	Autoencoder	VAE	DCGAN	Stable Diffusion
Sharpness and detail	●●●○○ 3/5	●●○○○ 2/5	●●●●○ 4/5	●●●●● 5/5
Ability to generate new images	●○○○○ 1/5	●●●○○ 3/5	●●●●○ 4/5	●●●●● 5/5
Training stability	●●●●● 5/5	●●●●○ 4/5	●●○○○ 2/5	●●●●○ 4/5
Generation speed	●●●●● 5/5	●●●●● 5/5	●●●●● 5/5	●●○○○ 2/5
Compute and memory efficiency	●●●●● 5/5	●●●●● 5/5	●●●●○ 4/5	●○○○○ 1/5
Control over the output (text)	●○○○○ 1/5	●○○○○ 1/5	●○○○○ 1/5	●●●●● 5/5

Key numbers from our runs

	Autoencoder	VAE	DCGAN	Stable Diffusion
Task	Reconstruct	Reconstruct + sample	Generate	Generate from text
Parameters	0.18 M	0.53 M	2.33 M (generator)	pretrained, very large
Mean MSE	0.0108	0.0145	n/a	n/a
Output size	28x28	28x28	28x28	512x512

When to use which

- **Autoencoder:** use it when you only need compact codes, denoising or anomaly scores.
- **VAE:** use it when you need both reconstruction and a smooth latent space to sample from.
- **DCGAN:** use it when you want fast, sharp samples and can tolerate unstable training.
- **Stable Diffusion:** use it when you want realistic, prompt-controlled images and have GPU time.

References (click a title to open the paper)

[1] Kingma, D. P., and Welling, M. (2014). Auto-Encoding Variational Bayes. ICLR.

[2] Goodfellow, I. et al. (2014). Generative Adversarial Nets. NeurIPS.

[3] Radford, A., Metz, L., and Chintala, S. (2016). Unsupervised Representation Learning with Deep Convolutional GANs. ICLR.

[4] Ho, J., Jain, A., and Abbeel, P. (2020). Denoising Diffusion Probabilistic Models. NeurIPS.

[5] Rombach, R. et al. (2022). High-Resolution Image Synthesis with Latent Diffusion Models. CVPR.